

Data Types

C is a statically typed language

- You must declare the type of every variable.
- Once declared, the type cannot be changed.
- This is because each type uses different memory and supports specific operations.

Types of Data

1. Primitive Data Types

- The most basic data types that are used for representing simple values.
- int, char, float, double, void

2. Derived Data Types

- Derived from the primitive or built-in datatypes.
- array, pointers, function

3. User Defined Data Types

- Defined by the user himself.
- structure, union, enum

Format Specifier

define the type of data being printed or read

Format Specifier

%c

%d, %i

%f

%ld or %li

%lf

Description

For character type.

For signed integer type.

For float type.

Long

Double

Format Specifier

%lu	Unsigned int or unsigned long
%lli or %lld	Long long
%llu	Unsigned long long
%o	Octal representation
%p	Pointer
%s	String
%u	Unsigned int

Escape Sequence

\t	-	Tab
\n	-	NewLine
\b	-	Backspace
\"	-	Double quotes
\'	-	Single quotes
\\	-	Backslash
\?	-	Question mark
%%	-	Percentage

Literal Constant

Literals are the constant values that are assigned to the variables.

- Integer Literal - Prefixes, Suffixes
- Float Literal
- Character Literal
- String Literal

1. Integer Literals

Integer literals are used to represent and store integer values.

Prefixes : They are classified based on their prefix (starting characters) which define the number system (base).

Types of Integer Literals:

1. Decimal (Base 10)

- Digits: 0-9
- No prefix
- Example: 56, 78

2. Octal (Base 8)

- Prefix: 0
- Digits: 0-7
- Example: 045, 076

3. Hexadecimal (Base 16)

- Prefix: 0x or 0X
- Digits: 0-9 and a-f or A-F
- Example: 0x23A, 0Xb4C

4. Binary (Base 2)

- Prefix: 0b or 0B
- Digits: 0 and 1
- Example: 0b101, 0B111

Suffixes: Integer literals can be written with suffixes to show their type.

12345678901234LL // long long int

500U // unsigned int

75L // long int

30UL // unsigned long int

100LL // long long int

200ULL // unsigned long long int